

Data Corruption Management (CPP-004)

CPP-Identifier	CPP-004
CPP-Label	Data Corruption Management
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Change history	Comments
Version 1.0.0 - 2025-08-29	Milestone version
Version 1.1.0 - 2026-03-26	Migration to XML, addition of Copy Management Policy as input for step 2.
Version 1.2.0 - 2026-05-18	Reassigned step numbers based on new numbering rules.

1. Description of the CPP

The TDA replaces damaged Files from replicated copies and reports on actions taken.

Inputs and outputs

Input(s)	
Data	AIP(s)
Metadata	Fixity metadata (checksums, algorithms and timestamps)
Documentation/ guidance	Copy management policy
Output(s)	
Data	AIP(s)
Metadata	Fixity metadata (checksums, algorithms and timestamps)
	Provenance metadata (event date, details and agents involved)
	Storage Management information (storage location)

Definition and scope

Data Corruption Management is a process, where a TDA restores corrupted *AIPs* from parallel copies. Corrupted *AIPs* are identified and flagged by the Integrity Checking (CPP-003) process, which periodically scans the fixity of *AIPs*.

When Integrity Checking (CPP-003) has flagged that an *AIP* is corrupted (i.e. that it has been unintentionally altered), the TDA recovers an intact copy from another storage medium in order to replace the corrupted *AIPs*. As part of the retrieval process, the checksums of the copied data are validated to verify a) the integrity of the data that is used as a source and, b) that the new target has been copied successfully. This is similar to the process of Replication (CPP-011). Subsequently, *Provenance metadata* is updated to maintain a record of the replacement procedure.

The replacing copy can be written to another storage medium than the corrupted one (e.g. in the case of magnetic tapes, where read-write operations to a single tape are kept to a minimum). In these cases, *Storage management information* must also be updated to reflect the new location of the copy.

The TDA may choose to replace the whole storage medium's content, triggering a process similar to Refreshment (CPP-030). This can be done when media-wide corruption or read/write errors are detected.

TData Corruption Management relies on the TDA having several parallel copies, preferably on different storage media and in different storage locations. The number of parallel copies, and their storage conditions are defined in the TDA's policies as maintained by Risk Mitigation (CPP-012) and a copy management policy in particular. In accordance with agreed on best-

practices, at least three copies are recommended, as there should exist at least two other valid copies in case one copy is corrupted or destroyed.

Process description

Trigger event(s)

Trigger Event	CPP-identifier
An AIP or File that has been flagged as corrupt	CPP-003 (Integrity Checking)
Media-wide corruption or read/write errors are detected	

Step-by-step description

No	Supplier	Input	Steps	Output	Customer
↓ sequence					
1	↺ alternative - Elaboration of a media- or AIP-specific inventory				
1.a	CPP-003 (Integrity Checking)	Integrity checking report	Identify and locate corrupted <i>AIPs</i>	Inventory of corrupted <i>AIPs</i> Storage medium with broken <i>AIPs</i>	
1.b		Report of media-wide errors	Identify and locate the <i>AIPs</i> on the corrupted medium.	Inventory of corrupted <i>AIPs</i> Storage medium with media-wide errors packages	
2	CPP-012 (Risk Mitigation)	Copy Management Policy	Select source medium to copy the <i>AIPs</i> from	Authoritative storage medium for replicating/ copying the <i>AIPs</i>	
3			Select target storage medium (can, and often is, be same as the original storage medium identified in step 1a)	Target storage medium	

No	Supplier	Input	Steps	Output	Customer
4			<i>Optional:</i> In case of media-wide errors: Provision of a fresh storage medium that will replace the old one.	The fresh storage medium that will replace the old one	
5		Source storage medium of <i>AIPs</i>	Gather AIP inventory and storage medium information to start the copy process (step 6)		
		Target storage medium			
		Inventory of <i>AIPs</i> involved in the process			
6	↓ sequence - For each AIP individually, copy AIP to storage medium				
6.1			Retrieve the <i>AIP</i> from the source storage medium		
6.2			Copy the <i>AIP</i> to the target storage medium	New copy of AIP	
6.3		Existing Fixity metadata	Validate the fixity of the <i>AIP</i> on the fresh storage medium	<i>Fixity Metadata</i>	
				<i>Optional:</i> Valid status (step 6.4)	
				<i>Optional:</i> Invalid status (go back to step 6.1)	
6.4			Update the fixity for the new <i>AIP</i> copy	<i>Fixity metadata</i>	

No		Supplier	Input	Steps	Output	Customer
6.5				<i>Optional:</i> If the target storage medium is different from the original storage medium with the broken <i>AIP</i> : Update the storage location for the new <i>AIP</i> copy	Storage management information	
7			Inventory of <i>AIPs</i> involved in the process	Check that all <i>AIPs</i> in the inventory have been successfully copied	<i>Optional:</i> Confirm completeness of the copy process (go to step 8)	
					<i>Optional:</i> Error (go back to copy process loop, step 6)	
8				<i>Optional:</i> In case of media-wide errors: Update information about the fresh storage medium (e.g. <i>File</i> locations, media identifiers) and mark the old medium and its contents as ready for deletion/decommissioning.	Storage management information	
9				Document the event and its timestamp	Provenance Metadata	
10			Original storage medium that has been refreshed	<i>Optional:</i> In case of media-wide errors Ensure data security and that confidentiality is not compromised by making sure that data on the original storage medium is properly deleted.		
11				Decommission the old storage medium	Record of decommissioning	

Rationale(s) and worst case(s)

Rationale	Impact of inaction or failure of the process
Parallel copies of the data.	Corrupted data cannot be restored unless parallel copies exist.
<i>Fixity metadata</i>	Data corruption cannot be detected, and replaced copies cannot be verified, without <i>Fixity metadata</i> .

2. Dependencies and relationships with other CPPs

Dependencies

CPP-ID	CPP-Title	Relationship description
CPP-003	Integrity Checking	Corrupted data is detected by periodic integrity checking.
CPP-012	Risk Mitigation	The number of parallel copies and how they are stored (media, locations) are defined in a TDA's policy that arises out of mitigating risks to preserved data..

Other relations

Relation	CPP-ID	CPP-Title	Relationship description
May require	CPP-005	Identifier Management	If a <i>File</i> is corrupted, it may need to be repaired or replaced. During this process, a new <i>PID</i> may be created.
Required by	CPP-013	Object Management Reporting	Fixing corrupted <i>AIPs</i> produces <i>Provenance metadata</i> and data for quality reporting to the stakeholders..
Affinity with	CPP-002	Checksum Validation	All new <i>AIP</i> copies must have their checksum validated to verify that the process was successful. The checksum validation is more mechanical in its nature in Data Corruption Management, only aiming at verification of the copy process. In contrast to CPP-002, it does not have to negotiate with producers or examine the results.
Affinity with	CPP-011	Replication	Corrupted copies are replaced by intact copies, effectively replicating the intact copy, but not creating a new parallel copy.
Not to be confused with	CPP-007	Virus Scanning	If a <i>File</i> is detected as infected and cannot be cleaned, it might be considered "damaged." However, CPP-004 typically applies to technical corruption or loss, rather than deliberately human-made damage such as malware-infected <i>Files</i> (CPP-007). In practice, infected <i>File</i> are more likely to be replaced (by the producer) or rejected.
Alternative to	CPP-027	File Repair	File Repair is an alternative (fallback) to Data Corruption Management in cases where no intact copy of corrupted or broken data is available, since repairing the structure of an altered copy is the only option.

3. Links to frameworks

Certification

Certification framework	Term used in framework to refer to the CPP	Section
CTS	“For each storage location, measures should be in place to ensure that unintentional or unauthorised changes can be detected and correct versions of data and metadata recovered”	R14 Storage and Integrity
Nestor Seal	“restoration of the archival information packages” “recovering archival information packages in the event of damage”	C15 Integrity: Functions of the archival storage
ISO 16363	“Recovery actions”	5.1.1.3.1 The repository shall record and report to its administration all incidents of data corruption or loss, and steps shall be taken to repair/replace corrupt or lost data.

Other frameworks and reference documents

Reference Document	Term used in framework to refer to the process	Section
OAIS	Disaster Recovery	4.2.3.4
PREMIS		/

4. Reference implementations

Use cases

Summary between 2021-2024 at CSC

Institutional background	
Institution	CSC – IT Center for Science Ltd., FI
Hyperlink	https://digitalpreservation.fi/en/services/quality_reports
Description	
Trigger event	2024a: Software error 2024b: Scheduled Integrity Checking 2022a: Human error 2022b: Scheduled Integrity Checking 2021: Scheduled Integrity Checking
Problem statement	2024a: Contents of one tape were lost 2024b: One corrupt AIP copy on a tape 2022a: One corrupt AIP copy on a disk 2022b: One corrupt AIP copy on a disk 2021: Ten corrupt AIP copies on a tape

Proposed solution	2024a: The tape was restored from other copies 2024b: A new copy was produced 2022a: A new copy was produced from the tape 2022b: The corrupted copy of the package had been unsuccessfully copied to storage during ingest earlier but had been later successfully copied to disk storage automatically. 2021: New copies were produced
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Publicly available documentation

Institution	Organisation type	Language	Hyperlink
TIB – Leibniz Information Centre for Science and Technology and University Library, DE	National library; Non-commercial digital preservation service; Research infrastructure; Research performing organisation	English	https://wiki.tib.eu/confluence/spaces/lza/pages/93608373/Archival+Storage#ArchivalStorage-Recovery
CSC – IT Center for Science Ltd., FI	Non-commercial digital preservation service	English	https://digitalpreservation.fi/en/services/quality_reports/2024 (section Quality Deviations Relating to Preserved Content in 2024)
		English	https://digitalpreservation.fi/en/services/quality_reports/2022 (Quality Deviations Related to the Data in Preservation in 2022)
Archivematica, CA	Digital preservation system	English	https://www.archivematica.org/en/docs/storage-service-0.23/recovery/#recovery